

Basic Exam — Spring 2010

Problem 1 [TODO]

Let $u_1, \dots, u_n$ be orthonormal basis of $\mathbb{R}^n$ and let $y_1, \dots, y_n$ be a collection of vectors in $\mathbb{R}^n$ satisfying $\sum_i \|y_i\|^2 < 1$. Prove that vectors $u_1 + y_1, \dots, u_n + y_n$ are linearly independent.

Solution. Consider $U = (u_1 \dots u_n)$. $\det(U) = 1$. Let $Y = (y_1 \dots y_n)$. Note that $\|Y\|_F^2 = \sum_i \|y_i\|^2 < 1$. We want to show that $\det(U + Y) \neq 0$. Note that all norms on $\text{Mat}_{n \times n}$ are equivalent and that $\det$ is a smooth function on this vector space.

Problem 2 [TODO]

Let A be $n \times n$ real symmetric matrix and let $\lambda_1 \geq \dots \geq \lambda_n$ be the eigenvalues of A . Prove that

$$\lambda_k = \max_{U, \dim U = k} \min_{x \in U, \|x\|=1} \langle Ax, x \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes the usual scalar product in $\mathbb{R}^n$ and the maximum is taken over all k dimensional subspaces of $\mathbb{R}^n$.

Solution. Let $v_1, \dots, v_n$ be the eigenvectors (and a orthonormal basis, spectral theorem) corresponding to eigenvalues $\lambda_1, \dots, \lambda_n$ respectively. Consider the function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ where

$$\begin{aligned} (a_1, \dots, a_k) &\rightarrow \left\langle A \sum_{i=1}^k a_i v_i, \sum_{i=1}^k a_i v_i \right\rangle \\ &= \left\langle \sum_{i=1}^k a_i \lambda_i v_i, \sum_{i=1}^k a_i v_i \right\rangle \\ &= \sum_{i=1}^k a_i^2 \lambda_i \end{aligned}$$

Note that

$$\frac{\partial f}{\partial a_i} = 2a_i \lambda_i.$$

We want to optimize f with respect to the condition that $\|a\|^2 = 1$. By the theory of lagrange multipliers it is that

$$2a_i \mu = 2a_i \lambda_i \Rightarrow a_i(\lambda_i - \mu) = 0$$

Thus $a_i = 0$ or $\lambda_i = \mu$. Not all $a_i = 0$ by our condition that $\|a\|^2 = 1$. Thus $\mu = \lambda_i$ for some $i \in \{1, \dots, k\}$. Note that $a_i^2 \mu = \lambda_i a_i^2$. Sum over all i we see

$$f(a) = \mu \sum_{i=1}^k a_i^2 = \mu.$$

Thus at any critical point our function equals the lagrange multiplier μ . Remember

$$a_i(\lambda_i - \mu) = 0 \quad \forall i \in \{1, \dots, k\}.$$

There must be at least one $a_i \neq 0$ since we are on the unit sphere. If a $a_i \neq 0$, then $\lambda_i = \mu$. Thus μ can be any eigenvalue of $\{\lambda_1, \dots, \lambda_k\}$. Since the possible critical values of f on the intersection of S^{n-1} with $\mathbb{R}(v_1, \dots, v_k)$ are the eigenvalues $\{\lambda_1, \dots, \lambda_k\}$, the absolute minimum of f on this domain is the the smallest of the critical values. Thus $\min_{v \in \mathbb{R}(v_1, \dots, v_k)} f(v) = \lambda_k$.

Problem 3

Let S and T be two normal transformations in the complex finite dimensional vector space V with a positive definite Hermitian inner product such that $ST = TS$. Prove that S and T have joint basis of eigenvectors.

Solution. Suppose S and T are two normal transformations then by the spectral theorem of normal matrices S and T are diagonalizable. Thus S, T are diagonalizable commuting matrices. Let $V_{\lambda_1}, \dots, V_{\lambda_n}$ be the eigenspaces of S . Then $\forall v \in V_{\lambda_i}$,

$$STv = TSv = \lambda_i Tv \Rightarrow Tv \in V_{\lambda_i}.$$

Hence V_{λ_i} are T invariant subspaces of V . Note that $m_{T|_{V_i}} \mid m_T$ hence if T is diagonalizable then m_T can be factored into linear factors hence so can $m_{T|_{V_i}}$ hence $T|_{V_i}$ is diagonalizable. Let $v_1^i, \dots, v_k^i$ be a eigenbasis for $T|_{V_i}$ (also a eigenbasis for $S|_{V_i}$). Put all basies for V_i together to get a basis of eigenvectors (of both S and T) for V .

Problem 4

- (a) Let $A = (a_{i,j})$ be $n \times n$ real symmetric matrix such that $\sum_{i,j} a_{i,j} x_i x_j \leq 0$ for every vector $(x_1, \dots, x_n)$ in $\mathbb{R}^n$. Prove that if $\text{tr}(A) = 0$ then $A = 0$.
- (b) Let T be a linear transformation in the complex finite dimensional vector space V with a positive definite Hermitian inner product. Suppose that $TT^* = 4T - 3I$, where I is identity transformation. Prove that T is positive definite Hermitian and find all possible eigenvalues of T .

Solution. (a): By spectral theorem, A is diagonalizable. $\sum_{i,j} a_{i,j} x_i x_j = x^T A x \leq 0$ for all $x \in \mathbb{R}^n$ then by definition A is a symmetric negative semi definite matrix. Hence all eigenvalues of A are nonpositive. If $\text{tr}(A) = 0$, that means all eigenvalues must be 0. Hence $A = 0$, since A is diagonalizable.

(b): $TT^* = (TT^*)^* = 4T^* - 3I = 4T - 3I \Rightarrow T^* = T$. Thus T is Hermitian and hence is diagonalizable with real eigenvalues.

$$\begin{aligned}
\langle Tv, Tw \rangle &= \langle v, T^*Tw \rangle \\
&= \langle v, TT^*w \rangle \\
&= \langle v, (4T - 3I)w \rangle \\
&= 4\langle v, Tw \rangle - 3\langle v, w \rangle
\end{aligned}$$

Let $v \in V$ be a nonzero eigenvector of T thus $Tv = \lambda v$. By the above calculation we see that

$$\begin{aligned}
\lambda^2 \langle v, v \rangle &= \langle Tv, Tv \rangle \\
&= 4\lambda \langle v, v \rangle - 3\langle v, v \rangle \\
(\lambda^2 - 4\lambda + 3) \langle v, v \rangle &= 0 \Leftrightarrow \lambda^2 - 4\lambda + 3 = 0
\end{aligned}$$

Thus

$$\lambda = \frac{3 \pm 2}{2} = \frac{5}{2}, \frac{1}{2}$$

Hence T is positive definite.

Problem 5

Let A, B be $n \times n$ complex matrices which have the same minimal polynomial $M(t)$ and the same characteristic polynomial $P(t) = (t - \lambda_1)^{a_1} \dots (t - \lambda_k)^{a_k}$, where $\lambda_i \neq \lambda_j$ for $i \neq j$. Prove that if $P(t)/M(t) = (t - \lambda_1) \dots (t - \lambda_k)$, then these matrices are similar.

Solution. Recall the following facts. Two matrices are similar if their invariant factors are the same. Furthermore, for a matrix A its invariant factors are polynomials $a_1, \dots, a_k \in \mathbb{C}[x]$ such that $a_i \mid a_{i+1}$, $a_k = m_A$ (minimal polynomial), and $p_A = \prod_{i=1}^k a_i$. Let $a_1, \dots, a_n$ be the invariant factors of A . Let $b_1, \dots, b_m$ be the invariant factors of B . Since

$$p_A = \prod_{i=1}^n a_i = \prod_{i=1}^m b_i = p_B$$

and that $\frac{p}{m_A} = \frac{p}{a_n} = \frac{p}{b_m} = \frac{p}{m_B}$ is a product of linear terms. We know that

$$\frac{p}{m_A} = \prod_{i=1}^{m-1} a_i = \prod_{i=1}^{n-1} b_i$$

Assume the contrary that a_{m-2} is isn't just a constant term 1. Then since $a_{m-2} \mid a_{m-1}$, $a_{m-1}a_{m-2}$ will have a squared linear term. But $\frac{p}{M}$ is a product of linear terms (square free), thus a_{m-2} and all previous terms must be 1. Thus A has only two non 1 invariant factors M and $\frac{p}{M}$. The same is true for B . Since both invariant factors are equal, A and B are similar.

Problem 6

Let $A = \begin{pmatrix} 4 & -4 \\ 1 & 0 \end{pmatrix}$.

- Find Jordan form J of A and a matrix P such that $P^{-1}AP = J$.
- Compute A^{100} and J^{100} .

(c) Find a formula for a_n , when $a_{n+1} = 4a_n - 4a_{n-1}$ and $a_0 = a$, $a_1 = b$.

Proof: (a):

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

(b):

$$J = (2I + E_{1,2})^{100} = (2I)^{100} + 100(2I)^{99}E_{ij} = \begin{pmatrix} 2^{100} & 100 * 2^{99} \\ 0 & 2^{100} \end{pmatrix} = 2^{99} \begin{pmatrix} 2 & 100 \\ 0 & 2 \end{pmatrix}$$

Hence

$$A^{100} = 2^{99}P \begin{pmatrix} 2 & 100 \\ 0 & 2 \end{pmatrix} P^{-1}$$

□

Problem 7 [TODO]

Let $\{f_n\}$ be a sequence of real-valued functions on the line, and assume that there is a $B < \infty$ such that $|f_n(x)| \leq B$ for all n and x . Prove that there is a subsequence $\{f_{n_k}\}$ such that $\lim_{k \rightarrow \infty} f_{n_k}(r)$ exists for all rational numbers r .

Solution. This is the diagonalization argument for Arzela Ascoli. Let $\mathbb{Q} = \{r_1, r_2, \dots\}$ be an enumeration of $\mathbb{Q}$. Note that $|f_n(r_1)| \leq B$. Thus by Bolzano Weirstrass there exists a subsequence $f_m^{(1)}$ of f_n such that

$$\lim_{m \rightarrow \infty} f_m^{(1)}(r_1)$$

is defined. We will proceed inductively. Suppose that for $f_n^{(k)}$ is a subsequence of f_n which converges pointwise on $\{r_1, \dots, r_k\}$. Then since $|f_n^{(k)}(r_{k+1})| \leq B$, there exists a subsequence of $f_n^{(k)}$, which we will denote $f_n^{(k+1)}$, such that

$$\lim_{n \rightarrow \infty} f_n^{(k+1)}(r_{k+1})$$

exists. Since $f_n^{(k+1)} \subseteq f_n^{(k)}$ it also converges pointwise for $\{r_1, \dots, r_k\}$. Thus $f_n^{(k+1)}$ is a subsequence of f_n which converges pointwise on $\{r_1, \dots, r_{k+1}\}$.

Thus $\forall k \in \mathbb{N}$, we have the following we have a subsequence $f_n^{(k)}$ of f_n such that

$$\lim_{n \rightarrow \infty} f_n^{(k)}(r_m)$$

exists for $m \in \{1, \dots, k\}$, furthermore we have that $f^{(k)} \subseteq f^{(k-1)} \subseteq \dots \subseteq f^{(1)} \subseteq f$. Now we will construct a subsequence of f_n , which we will call g_n where

$$g_n = f_n^{(n)}.$$

Note the following $\forall n \in \mathbb{N}$, the sequence $(g_{n+k})_{k=0}^{\infty}$ is a subsequence of $f^{(n)}$ by construction hence converges pointwise on $\{r_1, \dots, r_n\}$. Since this is true for all $n \in \mathbb{N}$, g_n is a subsequence of f_n which converges pointwise for all rationals.

Problem 8

Assume that K is a closed subset of a complete metric space (X, d) with the property that, for any $\varepsilon > 0$, K can be covered by a finite number of sets $B_\varepsilon(x)$, where $B_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}$. Prove that K is compact.

Solution. I am assuming that we need to prove that K is compact without using the fact that completeness and totally boundedness implies compactness. Since K is closed it contains all its limit points, thus for any cauchy sequence in K hence in X there exists a unique limit point in X which must also be in K . Thus K is complete. Let p_n be a sequence in K . By total boundedness, $\forall \varepsilon > 0$, there exists a $C_\varepsilon \subseteq K$ finite such that

$$K = \bigcup_{c \in C_\varepsilon} B_\varepsilon(c).$$

Consider C_1 , $\exists x \in C_1$ such that $B_1(x)$ contains a infinite number of elements of p_n . Denote such a subsequence $p_n^{(1)}$. Thus for all $m, n \in \mathbb{N}$:

$$d(p_n^{(1)}, p_m^{(1)}) < 2.$$

Proceed inductively suppose $p^{(k)}$ is a subsequence of $p^{(k-1)}$ such that $p_n^{(k)} \in \overline{B_{2^{-k+1}}}(x) \cap K$ for $x \in K$. $\overline{B_{2^{-k+1}}}(x) \cap K$ is closed and totally bounded hence there is a finite number of point $a_1, \dots, a_n$ such that

$$\overline{B_{2^{-k+1}}}(x) \cap K = \bigcup_{i=1}^n B_{2^{-k}}(x_i)$$

. There exists a x_i such that its ball contains a infinite number of $p^{(k)}$. Let $p^{(k+1)}$ denote such a subsequence. Hence $p^{(k+1)} \subset B_{2^{-k}}(x) \cap K$. Take the diagonal and then such a sequence is Cauchy hence convergent by completeness, hence there exists a convergent subsequence of p_n . Hence K is sequentially compact hence compact.

Problem 9 [TODO]

Assume that $f(x, y, z)$ is a real valued, continuously differentiable function such that $f(x_0, y_0, z_0) = 0$. If $\nabla f(x_0, y_0, z_0) \neq \vec{0}$, show that there is a differentiable surface, given parametrically by $(x(s, t), y(s, t), z(s, t))$ with $(x(0, 0), y(0, 0), z(0, 0)) = (x_0, y_0, z_0)$, on which $f = 0$.

Proof: This seems like a adaptation of the regular value theorem, where we replace C^∞ with C^1 . Let $v_0 = (x_0, y_0, z_0)$.

Since $df_{v_0} \neq 0$, we know there exists some change of basis of the coordinate around v_0 and around 0 such that $df_{v_0} = (1 \ 0 \ 0)$ in these coordinates. Ie the basis vectors $\mathbb{R}^3$ and $\mathbb{R}$ are

$$\begin{aligned}\partial_u &= a_1 \partial_x + a_2 \partial_y + a_3 \partial_z \\ \partial_v &= b_1 \partial_x + b_2 \partial_y + b_3 \partial_z \\ \partial_w &= c_1 \partial_x + c_2 \partial_y + c_3 \partial_z \\ \partial_t &= \lambda \partial_s\end{aligned}$$

where (x, y, z) are the coordinates on $\mathbb{R}^3$ and s is the coordinate on $\mathbb{R}$. Then let

$$\begin{aligned}u &= a_1 x + a_2 y + a_3 z \\ v &= b_1 x + b_2 y + b_3 z \\ w &= c_1 x + c_2 y + c_3 z \\ t &= \lambda s.\end{aligned}$$

You are simply applying invertible linear transformations on $\mathbb{R}^3$ and $\mathbb{R}$ which shouldn't affect the regularity of the function. Under these change of basis. $df_{v_0} = [1, 0, 0]$. Note that x, y, z are also functions of u, v, w . By the Implicit Function Theorem, $\exists$ a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ a C^1 function where $(s, t) \rightarrow (g(s, t), s, t)$ in our basis of (u, v, w) such that $f(g(s, t), s, t) = 0$ $(g(0, 0), 0, 0) = v_0$. Pull back along the invertible, linear - hence smooth change of coordinates to get $(x(s, t), y(s, t), z(s, t))$ in our original coordinates. Thus locally around $0 \in \mathbb{R}^2$, by the Implicit Function Theorem we can define a C^1 surface. $\square$

Problem 10 [TODO]

Let $f(x, y)$ be the function defined by

$$f(x, y) = \frac{xy}{\sqrt{x^2 + y^2}}$$

when $(x, y) \neq (0, 0)$ with $f(0, 0) = 0$.

- (a) Compute the directional derivatives of $f(x, y)$ at $(0, 0)$ in all directions where they exist. [TODO]
- (b) Is $f(x, y)$ differentiable at $(0, 0)$? Prove your answer. [TODO]

Proof: (a):

$$\begin{aligned}df_{(0,0)}\left(\begin{pmatrix} a \\ b \end{pmatrix}\right) &= \lim_{t \rightarrow 0} \frac{f(ta, tb) - f(0, 0)}{t} \\ &= \lim_{t \rightarrow 0} \frac{\frac{t^2 ab}{|t|}}{t} \\ &= \lim_{t \rightarrow 0} \left(\frac{t}{|t|}\right) ab = \lim_{t \rightarrow 0} \text{sgn}(t) ab \Rightarrow df_{0,0}\left(\begin{pmatrix} a \\ b \end{pmatrix}\right) \text{ exists} \Leftrightarrow ab = 0.\end{aligned}$$

□

Problem 11 [TODO]

Suppose $\sum_{n=1}^{\infty} |a_n| < \infty$. Let σ be a one-to-one mapping of $\mathbb{N}$ onto $\mathbb{N}$. The series $\sum_{n=1}^{\infty} a_{\sigma(n)}$ is called a “rearrangement” of $\sum_{n=1}^{\infty} a_n$. Prove that all rearrangements of $\sum_{n=1}^{\infty} a_n$ are convergent and have the same sum.

Problem 12 [TODO]

Assume that $\{f_n\}$ is a sequence of nonnegative continuous functions on $[0, 1]$ such that $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = 0$. Is it necessarily true that

- (a) There is a B such that $f_n(x) \leq B$ for $x \in [0, 1]$ for all n ? [TODO]
- (b) There are points x_0 in $[0, 1]$ such that $\lim_{n \rightarrow \infty} f_n(x_0) = 0$? [TODO]

Prove your answers.

Basic Exam — Fall 2011**Problem 13** [TODO]

Let (X, d) be a compact metric space and let $f : X \rightarrow X$ be a map satisfying

$$d(f(x), f(y)) < d(x, y), \quad \forall x, y \in X \text{ with } x \neq y.$$

Prove that there is a unique point $x \in X$ so that $f(x) = x$.

Solution.

Problem 14 [TODO]

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called *convex* if f satisfies

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \quad \forall x, y \in \mathbb{R}^n, \quad 0 \leq \alpha \leq 1.$$

Assume that f is continuously differentiable and that for some constant $c > 0$, the gradient ∇f satisfies

$$(\nabla f(x) - \nabla f(y)) \cdot (x - y) \geq c(x - y) \cdot (x - y), \quad \forall x, y \in \mathbb{R}^n,$$

where $\cdot$ denotes the dot product. Show that f is convex.

Solution.

Problem 15 [TODO]

Prove that the set of real numbers can be written as the union of uncountably many pairwise disjoint subsets, each of which is uncountable.

Solution.

Problem 16 [TODO]

If you rearrange the order of terms in a sum $\sum a_n$, sometimes you can change the limiting values. Find all the resulting limiting values of the following series. Prove your assertions.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ [TODO]

(b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ [TODO]

Solution. (a): $\forall \lambda \in \mathbb{R}$ we will show that we can rearrange

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

such that the resulting rearrangement equals λ .

Problem 17 [TODO]

Give an example of a function $f(x)$ on $[0, 1]$ with infinitely many discontinuities, but which is Riemann integrable. Include proof (don't just quote some theorem).

Solution. Let $a_n = \frac{1}{n}$. Consider the following function:

$$f(x) = \begin{cases} 1 & \text{if } x = \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

f has infinitely many discontinuities. Let $m_n = \frac{1}{2n(n-1)}$

$$P_n = \left\{ 0, \frac{1}{n} - m_n, \frac{1}{n} + m_n, \frac{1}{n-1} - m_n, \dots, \frac{1}{2} + m_n, 1 - m_n, 1 \right\}.$$

Let

$$G(I) = \sup_{x \in I} f(x) - \inf_{x \in I} f(x).$$

Note that $G([0, \frac{1}{n} - m_n]) = 1$, $G([\frac{1}{k} - m_n, \frac{1}{k} + m_n]) = 1$ for all k , but $G([\frac{1}{k} + m_n, \frac{1}{k} - m_n]) = 0$. Finally $G([1 - m_n, 1]) = 1$. Thus

$$\begin{aligned}
 R(P_n) &= U(f, P_n) - L(f, P_n) = \left(\frac{1}{n} - m_n \right) + 2(n-1)m_n + m_n \\
 &= \frac{1}{n} + 2(n-1)m_n = \frac{2}{n} \xrightarrow{n \rightarrow \infty} 0.
 \end{aligned}$$

Thus $\forall \varepsilon > 0$, we have that $\exists N \in \mathbb{N}$ such that for all $n \geq N$, $R(P_n) < \varepsilon$. We have more than proven the Riemann integrability of f .

Problem 18 [TODO]

Let f_n be a sequence of continuous functions on $[0, 1]$. Assume: (i) $f_n(x) \geq f_{n+1}(x)$ for all $x \in [0, 1]$; and (ii) $\lim_{n \rightarrow \infty} f_n(x) = 0$ for all $x \in [0, 1]$. Prove (don't just quote some theorem) that $f_n \rightarrow 0$ *uniformly* on $[0, 1]$.

Solution.

Problem 19 [TODO]

Let $f : \mathbb{R} \rightarrow M_{n \times n}$ be a continuous function, where $M_{n \times n}$ is the space of $n \times n$ matrices. Show that the function $g(t) = \text{rank}(f(t))$ is lower semi-continuous, meaning that if a sequence t_n converges to t then $g(t) \leq \liminf_n g(t_n)$. Is g always continuous?

Solution.

Problem 20 [TODO]

Assume that a complex matrix A satisfies $\ker((A - \lambda I)) = \ker((A - \lambda I)^2)$ for all $\lambda \in \mathbb{C}$. Show from first principles (i.e. without using the theory of canonical forms) that A must be diagonalizable.

Solution.

Problem 21 [TODO]

Let V be a finite dimensional inner product space, and let $L : V \rightarrow V$ be a self-adjoint linear operator. Let μ and ε be given. Suppose there is a unit vector $x \in V$ such that

$$\|L(x) - \mu x\| \leq \varepsilon.$$

Prove that L has an eigenvalue λ so that $|\lambda - \mu| \leq \varepsilon$.

Solution.

Problem 22 [TODO]

Let A be a 3×3 real matrix with $A^3 = I$. Show that A is similar to a matrix of the form

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$$

for some (real) θ . What values of θ are possible?

Solution.

Problem 23 [TODO]

- (a) State and prove the rank-nullity theorem. [TODO]
- (b) Suppose V , W , and U are finite dimensional vector spaces over $\mathbb{R}$ and that $T : V \rightarrow W$ and $S : W \rightarrow U$ are linear operators. Suppose further that T is one-to-one, S is onto, and $S \circ T = 0$. Prove that $\ker(S) \supseteq \text{image}(T)$ and that

$$-\dim(V) + \dim(W) - \dim(U) = \dim(\ker(S) / \text{image}(T)).$$

[TODO]

Solution.

Problem 24 [TODO]

Let A be an $m \times n$ real matrix, and let $b \in \mathbb{R}^m$. Suppose Ax and Ay are both of minimal distance to b (minimizing among members of $\text{image}(A)$). Prove that $x - y \in \ker(A)$.

Solution.

Basic Exam — Winter 2012**Problem 25** [TODO]

Let Ω denote the set of all closed subsets of $[0, 1]$ and let $\rho : \Omega \times \Omega \rightarrow [0, 1]$ be defined by

$$\rho(A, B) := \max \left\{ \sup_{x \in A} \inf_{y \in B} |x - y|, \sup_{y \in B} \inf_{x \in A} |x - y| \right\}.$$

Show that (Ω, ρ) is a metric space.

Solution.

Problem 26 [TODO]

Recall that $f : [a, b] \rightarrow \mathbb{R}$ is convex if for all $x, y \in [a, b]$ and $\alpha \in [0, 1]$, $f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$. Let $f_n : [a, b] \rightarrow \mathbb{R}$ be convex functions and suppose that $f(x) := \lim_{n \rightarrow \infty} f_n(x)$ exists at all $x \in [a, b]$ and is continuous on $[a, b]$. Prove that $f_n \rightarrow f$ uniformly.

Solution.

Problem 27 [TODO]

Prove the Bolzano–Weierstrass theorem in the following form: Each sequence $(a_n)_{n \in \mathbb{N}}$ of numbers a_n in the closed interval $[0, 1]$ has a convergent subsequence.

Solution.

Problem 28 [TODO]

For a sequence $\{a_n\}$ of non-negative numbers, let $s_n := \sum_{k=1}^n a_k$ and suppose that s_n tends to a number $s \in \mathbb{R}$ in Cesaro sense:

$$\lim_{n \rightarrow \infty} \frac{s_1 + \cdots + s_n}{n} = s.$$

Show that $\sum_{k=1}^{\infty} a_k$ exists and equals s .

Solution.

Problem 29 [TODO]

Prove that there is a unique continuous function $y : [0, 1] \rightarrow \mathbb{R}$ solving the equation

$$y(x) = e^x + \frac{y(x^2)}{2}, \quad x \in [0, 1].$$

Solution.

Problem 30 [TODO]

Let γ be a smooth curve from $(1, 0)$ to $(1, 0)$ in $\mathbb{R}^2 \setminus \{(0, 0)\}$ winding once around the origin in the clockwise direction. Compute the integral

$$I(\gamma) := \int_{\gamma} \frac{y \, dx - x \, dy}{x^2 + y^2}.$$

Solution.

Problem 31 [TODO]

Let $\mathbb{F}$ be the finite field of p elements, let V be a n -dimensional vector space over $\mathbb{F}$, and let $0 \leq k \leq n$. Compute the number of invertible linear maps $V \rightarrow V$.

Solution.

Problem 32 [TODO]

Let A be a $n \times n$ complex matrix. Prove that there are two sequences of matrices $\{B_i\}$ and $\{L_i\}$, such that L_i are diagonal with distinct eigenvalues, and $B_i L_i B_i^{-1} \rightarrow A$ as $i \rightarrow \infty$.

Solution.

Problem 33 [TODO]

Let $a_1 = 1$, $a_2 = 4$, $a_{n+2} = 4a_{n+1} - 3a_n$ for all $n \geq 1$. Find a 2×2 matrix A such that

$$A^n \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix}$$

for all $n \geq 1$. Compute the eigenvalues of A and use them to determine the limit $\lim_{n \rightarrow \infty} (a_n)^{1/n}$.

Solution.

Problem 34 [TODO]

Let A be a complex $n \times n$ matrix. State and prove under which conditions on A the following identity holds:

$$\det(e^A) = \exp(\operatorname{tr} A).$$

Here the matrix exponentiation is defined via the Taylor series $e^A = 1 + A + A^2/2! + A^3/3! + \dots$. You can assume known that this sum converges (entrywise) for all complex matrices A .

Solution.

Problem 35 [TODO]

- (a) Find a polynomial $P(x)$ of degree 2, such that $P(A) = 0$, for $A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}$. [TODO]
- (b) Prove that such $P(x)$ is unique, up to multiplication by a constant. [TODO]

*Solution.***Problem 36** [TODO]

Recall that the quadratic forms $Q_1(x, y)$ and $Q_2(x', y')$ are said to be equivalent if they are related by a non-singular change of coordinates $(x, y) \mapsto (x', y')$. Decide whether $Q_1 = xy$ and $Q_2 = x^2 + y^2$ are equivalent over $\mathbb{C}$ and whether they are equivalent over $\mathbb{R}$. If not, give a proof. If yes, find the matrix for change of coordinates.

*Solution.***Basic Exam — Fall 2012****Problem 37** [TODO]

Let $\{b_n\}_{n=1}^{\infty}$ be a sequence of real numbers with bounded partial sums, i.e., there is $M < \infty$ such that for all N , $|\sum_{n=1}^N b_n| \leq M$, and let $\{a_n\}_{n=1}^{\infty}$ be a sequence of positive numbers decreasing to 0. Prove the series $\sum a_n b_n$ converges.

*Solution.***Problem 38** [TODO]

Let $f(x)$ be a bounded real-valued function on the closed interval $[0, 1]$.

- (a) Give a (correct) definition of the Riemann integral $\int_0^1 f(x) dx$ that includes a necessary and sufficient condition for the integral to exist. [TODO]
- (b) Use your answer to (a) to prove that $\int_0^1 f(x) dx$ exists if f is non-decreasing (i.e. $f(x) \leq f(y)$ whenever $x \leq y$). [TODO]

*Solution.***Problem 39** [TODO]

Let $\{f_n(x)\}$ be a sequence of non-negative continuous functions on a compact metric space X . Assume $f_n(x) \geq f_{n+1}(x)$ for all n and x , so that $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ exists for every $x \in X$. Prove f is continuous if and only if f_n converges to f uniformly on X .

Solution.

Problem 40 [TODO]

A subset K of a metric space (X, d) is called *nowhere dense* if K has empty interior (i.e., if $U \subseteq K$, U open in X imply $U = \emptyset$). Prove the Baire theorem that if (X, d) is a complete metric space, then X is not a countable union of closed nowhere dense sets.

Solution.

Problem 41 [TODO]

A subset E of a metric space X is a G_δ set if $E = \bigcap_{n=1}^{\infty} G_n$ where each G_n is open in X . Use Problem 4 to prove that the set of rational numbers is *not* a G_δ subset of the set of real numbers.

Solution.

Problem 42 [TODO]

(a) Let $F(x, y)$ be a continuous function on the plane such that for every square S having its sides parallel to the axes, $\iint_S F(x, y) \, dx \, dy = 0$. Prove $F(x, y) = 0$ for all (x, y) . [TODO]

(b) Assume $f(x, y)$, $\frac{\partial f(x, y)}{\partial x}$, $\frac{\partial f(x, y)}{\partial y}$, $\frac{\partial}{\partial y} \left(\frac{\partial f(x, y)}{\partial x} \right)$ and $\frac{\partial}{\partial x} \left(\frac{\partial f(x, y)}{\partial y} \right)$ are all continuous in the plane. Use part (a) to prove that

$$\frac{\partial}{\partial y} \frac{\partial f(x, y)}{\partial x} = \frac{\partial}{\partial x} \frac{\partial f(x, y)}{\partial y}.$$

[TODO]

Solution.

Problem 43 [TODO]

Let A be an invertible $n \times n$ matrix with entries in $\mathbb{C}$. Suppose that the set of powers A^n of A , for $n \in \mathbb{Z}$, is bounded. Show that A is diagonalizable.

Solution.

Problem 44 [TODO]

Let H be an $n \times n$ Hermitian matrix with non-zero determinant. Use H to define a Hermitian form $[\cdot, \cdot]$ by the formula: for $x, y \in \mathbb{C}^n$ (column vectors!), $[x, y] = \{\}^t \bar{x} H y$, where the bar over x denotes complex conjugation and the t denotes transpose. Let W be a complex subspace of $\mathbb{C}^n$ such that $[w_1, w_2] = 0$ for all w_1 and w_2 in W . Show that $\dim W \leq n/2$. Give also for each n an example of an H for which $\dim W = n/2$ if n is even, or $\dim W = (n-1)/2$ if n is odd.

Solution.

Problem 45 [TODO]

Let A be an $m \times n$ real matrix with $m \geq n$. Let $b \in \mathbb{R}^m$. Let M be the set of vectors $x \in \mathbb{R}^n$ which minimize $|Ax - b|$. Show that $M = x_0 + N$ where N is the kernel of A , and x_0 is any element of M .

Solution.

Problem 46 [TODO]

Let A be a linear operator on a four dimensional complex vector space that satisfies the polynomial equation $P(A) = A^4 + 2A^3 - 2A - I = 0$, where I is the identity operator. Let $B = A + I$ and suppose $\dim(\text{range}(B)) = 2$. Finally, suppose that $|\text{Tr}(A)| = 2$. Give a Jordan canonical form of A .

Solution.

Problem 47 [TODO]

Show that an $n \times n$ matrix A can be factored as $A = LU$, where L is lower triangular and U is upper triangular, provided each determinant $\det A_j$, for $j = 1, \dots, n-1$, is non-zero, where A_j is the submatrix of A consisting of the first j rows and the first j columns.

Solution.

Problem 48 [TODO]

Let M be an $n \times m$ matrix. Prove that the row rank of M equals the column rank of M . Also, however you prove it, interpret this result as an equality of the dimensions of two vector spaces naturally attached to the map defined by M .

Solution.